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LLM Evals 101

Testing the quality of your AI application

Measure whether your application does what you built it to do.



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THE PURPOSE

Why evals matter

An application can run successfully and still give the wrong answer.

01 The same input can produce different responses.

02 Different wording can still be correct.

03 Changes can break behavior that previously worked.

REQUEST

Suggest a vegetarian dinner.

RESPONSE

Try a lemon herb chicken with spinach.

✕ FAIL Not vegetarian — yet no error was raised

Unit tests
Application logic and structure

Evals
Meaning and behavior

Model benchmarks measure general capability. Application evals test your specific task.

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QUALITY CRITERIA

What good looks like

Correctness
Accurate facts and task results.

Faithfulness
Claims supported by supplied context — which may itself contain errors.

Relevance
An answer that addresses the request.

Instruction following
Required constraints and format.

Uncertainty
Clarification when information is missing.

Safety
Appropriate handling of harmful requests and sensitive information.

EXAMPLE Our recipe should be vegetarian, use the requested ingredients, and fit the stated time limit.

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THE BUILDING BLOCKS

Anatomy of an eval

01
Test case

Input, context, expected behavior.

→

02
Application output

What the application produced.

→

03
Evaluator

The check that grades it.

→

04
Result

Score or label, plus a reason.

Reference-based

Compare with a known answer or expected facts.

Reference-free

Assess against criteria without an ideal answer. Can still use source context and a rubric.

EXAMPLE

Judge whether the suggested recipe satisfies the constraints, even if its wording differs from a reference.

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TEST COVERAGE

Evaluation datasets

A golden dataset contains curated inputs and trusted expectations.

CASE CATEGORIES

Common requests

Boundary cases

Missing or ambiguous information

Unsafe or adversarial requests

Previously observed failures

COMMON	Vegetarian dinner within 30 minutes
BOUNDARY	Recipe listed at 30 minutes versus 31
AMBIGUOUS	“Make something suitable for my diet”

Start with representative cases. Expand coverage as you discover failures.

Review AI-generated test cases before trusting them.

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Evaluation methods

01

Code / rules

Check objective requirements.

EXAMPLE

Required fields exist in a JSON response.

02

LLM-as-judge

Assess meaning against a rubric.

EXAMPLE

The recipe follows the user's constraints.

03

Human review

Assess nuance and calibrate automated checks.

EXAMPLE

Review unclear or disputed results.

Combine methods according to the task.

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AUTOMATED JUDGMENT

LLM-as-judge

A useful judge needs explicit criteria.

`Question` + `Response` + `Relevant context` + `Rubric` → `Judge` → `Pass / Fail + short reason`

EXAMPLE RUBRIC

Pass if the recipe:

- [ ] Is vegetarian.
- [ ] Includes tomatoes and spinach.
- [ ] Meets the 30-minute limit using the supplied timing.
- [ ] Makes no unsupported recipe claims.

Check judge decisions against human-reviewed examples.

Judges can make mistakes and show preferences for answer length or position.

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Scores and acceptance criteria



Pass / Fail

Did it meet a specific requirement?



Numeric score

How well did it meet defined rating levels?



Pairwise comparison

Which response is better, or are they tied?

— Review individual failures and category results.

— Define acceptance criteria for your application's risks.

— Track latency and cost alongside quality.

A strong overall score can hide an important failure.

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COMPARING VERSIONS

Experiments and regressions

Experiment
Compare application versions.

Regression
Previously working behavior now fails.

KEEP FIXED

Same test cases

Same evaluator and criteria

Recorded prompt, model, dataset, and evaluator versions

VERSION A

✓ Respects the vegetarian constraint.

VERSION B

✓ Produces shorter answers.

✗ Sometimes recommends meat.

REGRESSION

Inspect both improvements and regressions before releasing a change.

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RAG evaluation

Evaluate both the retrieved information and the generated answer.

01 · RETRIEVAL

Coverage / recall
Did we retrieve the information needed?

Relevance / precision
How much retrieved content was useful?

→

02 · ANSWER

Faithfulness
Does the answer follow the retrieved material?

Correctness and relevance
Does it accurately answer the request?

EXAMPLE

The user asks for vegetarian dinner recipes, but retrieval returns dessert recipes.

Component checks locate failures. End-to-end checks assess the user's result.

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Agent evaluation

Outcome
Did the task actually succeed?

Tool use
Correct tools and valid arguments.

Process
Required checks and permissions.

Conversation
Constraints preserved across turns.

ASSISTANT SAYS

"Ingredients added to your basket."

BASKET CONTAINS

Pasta	✓
Tomatoes	✓
Spinach — missing	✗

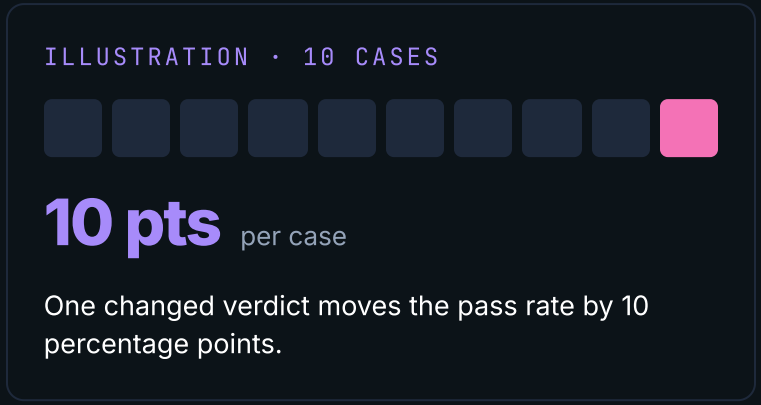
✗ FAIL The message must match the actual basket.

Different valid action sequences can reach the same successful outcome.

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Trustworthy evaluation

Held-out cases	Use unseen examples for final checks.
Repeated runs	Check whether important behavior remains reliable.
Sample size	Interpret results from small datasets carefully.
Versioned evaluation	Record changes to cases, criteria, and evaluators.



Consistency means dependable behavior, not identical wording.

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THE CONTINUOUS LOOP

Offline and online evaluation

Offline
Run controlled datasets before release.

Online
Assess real production interactions.



Observability records what happened.
Evaluation assesses whether it met expectations.

Begin with one important failure and build coverage from there.